Eric **Bridgeford**

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Education

09/18 – 08/23 **Ph.D.** Johns Hopkins University, Department of Biostatistics — Baltimore, MD
Principal Investigators: Joshua Vogelstein, Brian Caffo
Work focused on measurement error and measurement replicability; particularly, optimal strategies for maximizing replicability and mitigating batch effects in multivariate data
Coursework Includes: Probability Theory, Statistical Theory, Advanced Data Science, Bayesian Statistics, Graph Statistics, Advanced Statistical Methods, Approximation Theory, Causal Inference, Financial Derivatives

09/13 – 05/17 **B.S.** Johns Hopkins University, Biomedical Engineering and Computer Science — Baltimore, MD
Thesis work supervised by Dr. Joshua T. Vogelstein on project entitled:
Functional Neurodata Graphs Service: a One-Click Pipeline for the Reliable Estimation of Functional Connectomes.
Coursework Includes: High Dimensional Approximation, Statistical Learning and Signal Processing, Machine Learning, Probability and Statistics, Discrete Mathematics, Database Systems, Algorithm and Computation Theory, Systems Bioengineering, Object Oriented Software Engineering, Data Structures

Experience

Academic

10/24 – now **Johns Hopkins University, Applied Biomedical Engineering** — Baltimore, MD
Lecturer
Teaching advanced graph analytics to industry professionals in classes of 10-15 students. Developed curriculum that builds theory-driven intuition and enables students to translate complex statistical methods to their applied contexts, resulting in improved practical implementation.

01/24 – now **Stanford University, Poldrack Lab** — Stanford, CA
Postdoctoral Scholar with Russell Poldrack
Addressing critical challenges in neuroimaging by developing statistical frameworks that reduce selection biases and measurement errors. Create innovative methodologies that enhance validity and reliability of brain imaging research findings.

10/17 – 01/24 **Johns Hopkins University, Center for Imaging Science** — Baltimore, MD
Postdoctoral Scholar with Joshua T. Vogelstein
Graduate Research Assistant with Joshua T. Vogelstein and Brian Caffo
Identified reliability issues in high-dimensional neuroimaging data affecting research outcomes. Led projects to quantify variability sources and develop robust hypothesis testing. Implemented dimensionality reduction techniques that improve statistical power and enhanced result interpretability across multiple studies. Department representative for mentorship of data analytics-focused capstones and statistical consultation for lab mates.

08/14 – 05/17 **Center for Imaging Science, Johns Hopkins University** — Baltimore, MD
Undergraduate Researcher under Joshua T. Vogelstein
Design and implementation of an open-source, cloud-deployable fMRI and DWI pipeline for robust one-click analysis.

05/14 – 02/16 **Complex Systems Group, University of Pennsylvania** — Philadelphia, PA
Undergraduate Researcher under Danielle S. Bassett
Assisted in the development of novel network theory statistic and code package to compare the small worldness of weighted, real world networks.

Professional

06/25 – 08/25 **Scale.ai** — San Francisco, CA
Human Frontier Collective Specialist
Summer internship providing tailored domain expertise for training, evaluating, and annotating cutting-edge datasets for large language models.

01/18 – 10/18 **Atana, Inc.** Baltimore, MD
Vice President of Engineering
Led software development team during company growth phase, shaping technological strategy. Managed development of MVPs of core product, healthcare pilot programs, and coauthored technical documentation, which contributed to securing $500,000 in funding before departing for graduate studies.

Teaching

01/22 – now **Whiting School of Engineering, Johns Hopkins University** Baltimore, MD
Provide lectures on network-valued data and applications to connectomics as part of Introduction to Computational Medicine undergraduate course.

08/23 **Joint Statistical Meetings (JSM) Continuing Education Course** Toronto, CA
Taught 4 hour continuing education course with Jaewon Chung on introducing the basics of network data, statistical network modeling, and statistical analysis of network data. Included hands-on instruction in `python` programming language.

01/21 – 06/23 **School of Public Health, Johns Hopkins University** Baltimore, MD
Teaching Assistant for Master's Capstone Projects under Dr. Marie Diener-West. Responsibilities include end-to-end project advisement and project scoping on a weekly basis with 8-10 students for about 30 minutes per meeting over the course of the semester.

08/19 – 06/23 **Department of Biostatistics, Johns Hopkins University** Baltimore, MD
Teaching assistant for Statistical Methods I, II, III, and IV. Responsibilities include grading and office hours.

03/22 – 05/22 **ABCD-ReproNim Course** online
Provided lectures on unsupervised machine learning with network data with applications to Human Connectomes.

08/20 – 12/21 **Department of Biostatistics, Johns Hopkins University** Baltimore, MD
Teaching Assistant for Advanced Data Science I/II, under Drs. Jeff Leek and Roger Peng. Responsibilities include grading, class design, and leading breakout sessions for class meetings.

08/17 – 05/18 **Biomedical Engineering Department, Johns Hopkins University** Baltimore, MD
Teaching Assistant for 580.437/697 Neuro Data Design 1 under Dr. Joshua Vogelstein. Responsibilities include end-to-end project advisement, project scoping, and grading for about 20 students, with weekly meetings totaling about 5 hours.

01/17 – 05/17 **Computer Science Department, Johns Hopkins University** Baltimore, MD
Course Assistant for 600.475 Introduction to Machine Learning under Dr. Raman Arora.

Organizations and Volunteer Work

03/08 – 05/17 **Special Olympics Male Gymnastics Coach, Hatboro YMCA** Hatboro, PA
Volunteer work mentoring & coaching special needs gymnasts. Head male gymnastics coach from 03/11 – 05/14.

04/14 – 05/17 **Sigma Chi Fraternity, $K\Upsilon$ Chapter** JHU, Baltimore, MD
Chapter Risk manager from 09/14 – 05/15.

Awards

02/18 **b0x Sponsor Prize** ETHDenver, Denver, CO
developed Delphi, a framework for a decentralized oracle service between Hyperledger Fabric private networks and the Ethereum public network.

09/17 **JP Morgan and Booz Allen Hamilton Sponsor Prizes** JHU Hophacks, Baltimore, MD
developed Illuminate, an application that empowers verifiable, frictionless donations for worldwide disaster relief by leveraging the Ethereum public network at Hop Hacks.

05/17 **CS Departmental Honors with Thesis** JHU, Baltimore, MD
awarded for maintaining a cumulative GPA of 3.5 or higher within courses specific to the computer science department and acceptance of senior research thesis.

05/17 **BME Departmental Honors** JHU, Baltimore, MD
awarded for maintaining a cumulative GPA of 3.5 or higher within courses specific to the Biomedical Engineering department.

05/17 **General Honors** JHU, Baltimore, MD
awarded for maintaining cumulative GPA of 3.5 or higher.

09/14 – 05/17 **Martha A. Laverty Scholar** JHU, Baltimore, MD
Grant awarded for merit achievement.

05/15 – 05/17 **Dean's List** JHU, Baltimore, MD
Awarded for maintaining a GPA above a 3.5/4.0.

09/15 **Pennapps Everyblock Sponsor Prize** UPenn, Philadelphia, PA
Awarded for development of Strollsafe, an application that enables users to seamlessly view crime risk for a given area while maneuvering Philadelphia at University of Pennsylvania's PennApps.

Skills

- Computational: Python, R, SQL, Java, C++, Docker, Jupyter, PyTorch, LaTeX, git, unit testing, CI/CD, visualization.
- Statistical: Causal inference, hypothesis testing, graph theory, non-parametric statistics, regression, anomaly detection, Bayesian modeling, dimensionality reduction, reproducibility, experimental design.
- Technical: object-oriented design, API engineering, high-performance/cloud computing, pipeline engineering, big data.
- Cross-Domain: philosophy of science, domain translation, scientific communication, interdisciplinary research.

Publications

Books and Book Chapters

1. **Eric W. Bridgeford**, Alexander Loftus, and Joshua T. Vogelstein. *Hands on Network Machine Learning with Python*. Upcoming publication with Cambridge University Press, 2025.
2. **Eric W. Bridgeford**, Daniel Sussman, Vince Lyzinski, Yichen Qin, Youngser Park, Brian Caffo, Carey Priebe, and Joshua T. Vogelstein. *What Is Connectome Coding?* Ed. by E Boyden and K Chung. In: Functional, Structural, and Molecular Imaging, and Big Data Analysis (edited by E. Boyden and K. Chung) pp. 63–74, 2018, pp. 63–74.

Articles Accepted to Peer-Reviewed Journals

1. **Eric W. Bridgeford**, Michael Powell, Gregory Kiar, Stephanie Noble, Jaewon Chung, Sambit Panda, Ross Lawrence, Ting Xu, Michael Milham, Brian Caffo, and Joshua T. Vogelstein. "When no answer is better than a wrong answer: A causal perspective on batch effects". *Imaging Neuroscience* 3 (Mar. 2025).
2. Vivek Gopalakrishnan, Jaewon Chung, **Eric W. Bridgeford**, Benjamin D. Pedigo, Jesús Arroyo, Lucy Upchurch, G. Allan Johnson, Nian Wang, Youngser Park, Carey E. Priebe, and Joshua T. Vogelstein. "Multiscale comparative connectomics". *Imaging Neuroscience* 3 (May 2025).
3. Alexandra Maertens, Breanne Kincaid, **Eric W. Bridgeford**, Celine Brochot, Arthur de Carvalho E Silva, Jean-Lou C M Dorne, Liesbet Geris, Trine Husøy, Nicole Kleinstreuer, Luiz Carlos Maia Ladeira, Alistair Middleton, Joe Reynolds, Blanca Rodriguez, Erwin L Roggen, Giulia Russo, Kris Thayer, and Thomas Hartung. "From cellular perturbation to probabilistic risk assessments". English. *ALTEX: Alternatives to Animal Experimentation* 42.3 (2025). ISSN: 1868-596X.
4. Zeyi Wang, **Eric W. Bridgeford**, Shangsi Wang, Joshua T. Vogelstein, and Brian Caffo. "Statistical Analysis of Data Repeatability Measures". *Accepted at International Statistical Review* (June 2024).
5. Benjamin D. Pedigo, Mike Powell, **Eric W. Bridgeford**, Michael Winding, Carey E. Priebe, and Joshua T. Vogelstein. "Generative network modeling reveals quantitative definitions of bilateral symmetry exhibited by a whole insect brain connectome". *eLife* (Mar. 2023).
6. Ting Xu, Gregory Kiar, Jae Wook Cho, **Eric W. Bridgeford**, Aki Nikolaidis, Joshua T. Vogelstein, and Michael P. Milham. "ReX: an integrative tool for quantifying and optimizing measurement reliability for the study of individual differences". *Nature Methods* 20 (June 2023), pp. 1025–1028. ISSN: 1548-7105.

7. Jaewon Chung, **Eric W. Bridgeford**, Jess Arroyo, Benjamin D. Pedigo, Ali Saad-Eldin, Vivek Gopalakrishnan, Liang Xiang, Carey E. Priebe, and Joshua T. Vogelstein. "Statistical Connectomics". *Annu. Rev. Stat. Appl.* 8.1 (Mar. 2021), pp. 463–492. ISSN: 2326-8298.
8. **Eric W. Bridgeford**, Shangsi Wang, Zhi Yang, Zeyi Wang, Ting Xu, Cameron Craddock, Jayanta Dey, Gregory Kiar, William Gray-Roncal, Carlo Colantuoni, Christopher Douville, Stephanie Noble, Carey E. Priebe, Brian Caffo, Michael Milham, Xi-Nian Zuo, Consortium for Reliability and Reproducibility, and Joshua T. Vogelstein. "Eliminating accidental deviations to minimize generalization error and maximize replicability: applications in connectomics and genomics". *PLoS Computational Biology* 17.9 (Sept. 2021).
9. Donniell E. Fishkind, Felix Parker, Hamilton Sawczuk, Lingyao Meng, **Eric W. Bridgeford**, Avanti Athreya, Carey Priebe, and Vince Lyzinski. "The phantom alignment strength conjecture: practical use of graph matching alignment strength to indicate a meaningful graph match". *Appl. Network Sci.* 6.62 (Dec. 2021), pp. 1–27. ISSN: 2364-8228.
10. Ross M. Lawrence, **Eric W. Bridgeford**, Patrick E. Myers, Ganesh C. Arvapalli, Sandhya C. Ramachandran, Derek A. Pisner, Paige F. Frank, Allison D. Lemmer, Aki Nikolaidis, and Joshua T. Vogelstein. "Standardizing human brain parcellations". *Sci. Data* 8.78 (Mar. 2021), pp. 1–9. ISSN: 2052-4463.
11. Roger Peng, Athena Chen, **Eric W. Bridgeford**, Jeff Leek, and Stephanie Hicks. "Diagnosing Data Analytic Problems in the Classroom". *Journal of Statistics and Data Science Education* (Aug. 2021), pp. 267–276.
12. Joshua T. Vogelstein†, **Eric W. Bridgeford**†, Minh Tang, Da Zheng, Christopher Douville, Randal Burns, and Mauro Maggioni. "Supervised dimensionality reduction for big data". *Nat. Commun.* 12.2872 (May 2021), pp. 1–9. ISSN: 2041-1723.
13. Jordan Yoder, Li Chen, Henry Pao, **Eric W. Bridgeford**, Keith Levin, Donniell E. Fishkind, Carey Priebe, and Vince Lyzinski. "Vertex nomination: The canonical sampling and the extended spectral nomination schemes". *Comput. Statist. Data Anal.* 145 (May 2020), p. 106916. ISSN: 0167-9473.
14. Jaewon Chung, Benjamin D. Pedigo, **Eric W. Bridgeford**, Bijan K. Varjavand, Hayden S. Helm, and Joshua T. Vogelstein. "GraSPy: Graph Statistics in Python". *Journal of Machine Learning Research (JMLR)* 20.158 (Jan. 2019).
15. Carey E. Priebe, Youngser Park, Joshua T. Vogelstein, John M. Conroy, Vince Lyzinski, Minh Tang, Avanti Athreya, Joshua Cape, and **Eric W. Bridgeford**. "On a two-truths phenomenon in spectral graph clustering". *Proc. Natl. Acad. Sci. U.S.A.* 116.13 (Mar. 2019), pp. 5995–6000. ISSN: 0027-8424.
16. Joshua T. Vogelstein, **Eric W. Bridgeford**, Benjamin D. Pedigo, Jaewon Chung, Keith Levin, Brett Mensh, and Carey E. Priebe. "Connectal coding: discovering the structures linking cognitive phenotypes to individual histories". *Curr. Opin. Neurobiol.* 55 (Apr. 2019), pp. 199–212. ISSN: 0959-4388.
17. Joshua T. Vogelstein, **Eric W. Bridgeford**, Qing Wang, Carey E. Priebe, Mauro Maggioni, and Cencheng Shen. "Discovering and deciphering relationships across disparate data modalities". *eLife* (Jan. 2019).
18. Joshua T. Vogelstein, Eric Perlman, Benjamin Falk, Alex Baden, William Gray Roncal, Vikram Chandrashekhar, Forrest Collman, Sharmishtaa Seshamani, Jesse L. Patsolic, Kunal Lillaney, Michael Kazhdan, Robert Hider, Derek Pryor, Jordan Matelsky, Timothy Gion, Priya Manavalan, Brock Wester, Mark Chevillet, Eric T. Trautman, Khaled Khairy, **Eric W. Bridgeford**, Dean M. Kleissas, Daniel J. Tward, Ailey K. Crow, Brian Hsueh, Matthew A. Wright, Michael I. Miller, Stephen J. Smith, R. Jacob Vogelstein, Karl Deisseroth, and Randal Burns. "A community-developed open-source computational ecosystem for big neuro data". *Nat. Methods* 15.11 (Nov. 2018), pp. 846–847. ISSN: 1548-7105.
19. Sarah Feldt Muldoon, **Eric W. Bridgeford**, and Danielle S. Bassett. "Small-World Propensity and Weighted Brain Networks". *Sci. Rep.* 6.22057 (Feb. 2016), pp. 1–13. ISSN: 2045-2322.

In progress

1. **Eric W. Bridgeford**. "Measure Theoretic Probability Theory: A Hands-On, Proof-Based Introduction". *In preparation* (2025).
2. **Eric W. Bridgeford**, Brian Caffo, Maya B. Mathur, and Russell A. Poldrack. "How causal perspectives can inform problems in computational neuroscience". *Under review at Nature Neuroscience* (Jan. 2025).
3. **Eric W. Bridgeford**, Jaewon Chung, Robert J. Anderson, Ali Mahzarnia, Jacques A. Stout, Hae Sol Moon, Zay Yar Han, Joshua T. Vogelstein, and Alexandra Badea. "Network Biomarkers of Alzheimer's Disease Risk Derived from Joint Volume and Texture Covariance Patterns in Mouse Models". *Under review at PLoS One* (Feb. 2025), p. 2025.02.05.636582.
4. Jaewon Chung, **Eric W. Bridgeford**, Michael Powell, Derek Pisner, Ting Xu, and Joshua T. Vogelstein. "The Heritability of Human Connectomes: a Causal Modeling Analysis". *bioRxiv, Under internal revision* (Nov. 2023).
5. **Eric W. Bridgeford**, Jaewon Chung, Brian Gilbert, Sambit Panda, Ashwin DeSilva, Censheng Shen, Brian Caffo, and Joshua T. Vogelstein. "Learning sources of variability from high-dimensional observational studies". *In preparation* (2023).
6. Ross Lawrence, Alex Loftus, Gregory Kiar, **Eric W. Bridgeford**, William Gray Roncal, Vikram Chandrashekhar, Disa Mhembere, Sephira Ryman, Xi-Nian Zuo, Daniel S. Margulies, R. Cameron Craddock, Carey E. Priebe, Rex Jung, Vince D. Calhoun, Brian Caffo, Randal Burns, Michael P. Milham, Joshua T. Vogelstein, and Consortium for Reliability and Reproducibility (CoRR). "A low-resource reliable pipeline to democratize multi-modal connectome estimation and analysis". *bioRxiv* vol. 8.78 (Nov. 2021).

† *co-first author*

7. Sambit Panda, Satish Palaniappan, Junhao Xiong, **Eric W. Bridgeford**, Ronak Mehta, Cencheng Shen, and Joshua T. Vogelstein. "hyppo: A Multivariate Hypothesis Testing Python Package". *arXiv* (July 2019).

Articles Accepted to Peer-Reviewed Conferences

1. Qingyang Wang, Michael A. Powell, Ali Geisa, **Eric W. Bridgeford**, Carey E. Priebe, and Joshua T. Vogelstein. *Why do networks have inhibitory/negative connections?* *IEEE/CVF International Conference on Computer Vision (ICCV)* (Oct. 2023), pp. 01–06.
2. Qingyang Wang, Michael Alan Powell, **Eric W. Bridgeford**, Ali Geisa, and Joshua T. Vogelstein. *Polarity Is All You Need to Learn and Transfer Faster*. *ICML* (2023).

Conference Posters

1. **Eric W. Bridgeford**, Maya Mathur, Brian Caffo, and Russell Poldrack. "How causal perspectives inform problems in computational neuroscience". *Organization for Human Brain Mapping*. June 2025.
2. **Eric W. Bridgeford** et al. "Batch Effects are Causal Effects: Applications in Connectomics". *Organization for Human Brain Mapping*. June 2022.
3. Gregory Kiar, William R Gray Roncal, Disa Mhembere, **Eric W. Bridgeford**, Shan gsi Wang, Carey Priebe, Randal Burns, and Joshua T Vogelstein. "A Principled Approach to Statistical Connectomics and Mega-Analysis". *Organization for Human Brain Mapping (OHBM)*. June 2018.
4. **Eric W. Bridgeford**, Gregory Kiar, Will Gray Roncal, Disa Mehembre, Randal Burns, and Joshua T Vogelstein. "MRImages to Graphs: A One Click Community Pipeline for MR Connectome Analysis". *Institute for Computational Medicine Poster Session*. 2015.
5. Gregory Kiar and et al. "Community Connectomics via Cloud Computing Utilizing m2g - a Reference Pipeline". *Organization for Human Brain Mapping (OHBM)*. 2015.
6. Sarah Muldoon, **Eric W. Bridgeford**, and Danielle Bassett. "Quantifying Small Worldness in Weighted Brain Networks: Small-World Propensity". *Society for Neuroscience (SfN)*. Oct. 2015.
7. Joshua T. Vogelstein and et al. "The Open Connectome Project & NeuroData: Enabling Data Driven Neuroscience at Scale". *Society for Neuroscience (SfN)*. Oct. 2015.

Talks and Invited Lectures

1. **Eric W. Bridgeford**. "An Introduction to Network-Valued Data and Network Analysis". Introduction to Computational Medicine, 2023.
2. **Eric W. Bridgeford** and Jaewon Chung. "Hands on Network Machine Learning". Joint Statistical Meetings Continuing Education Course, 2023.
3. **Eric W. Bridgeford**. "An Introduction to Network-Valued Data and Network Analysis". Introduction to Computational Medicine, 2022.
4. **Eric W. Bridgeford**. "Hands on Network Machine Learning with Scikit-Learn and Graspologic". Brain Informatics PI Meeting, 2022.
5. **Eric W. Bridgeford**. "Hands on Network Machine Learning with Scikit-Learn and Graspologic". Berlin Connectomics Meeting, 2022.
6. **Eric W. Bridgeford**. "Hands on Network Machine Learning with Scikit-Learn and Graspologic". Institute of Computational Medicine Night, 2022.
7. **Eric W. Bridgeford**. "Unsupervised Machine Learning with Connectomics Data". ABCD Reproducible Neuroimaging Course, 2022.
8. **Eric W. Bridgeford**. "A Principled Approach to Statistical Connectomics and Mega-Analysis". Organization of Human Brain Mapping, 2018.
9. Joshua T Vogelstein and **Eric W. Bridgeford**. "Cross Modality Connectome Properties". Society for Neuroscience, 2017.
10. **Eric W. Bridgeford**. "From the Functional Brain to the Connectome: An Introduction to Neuroscience Research in the 21st Century". JHU Splash, 2016.

Software Packages

1. **Eric W. Bridgeford**, Michael Powell, Brian Caffo, and Joshua T. Vogelstein. "Causal Batch Effects". *R package*. 2024.
2. Derek Pisner, Alex Loftus, Gregory Kiar[†], **Eric Bridgeford**[†], Will Gray Roncal, and Joshua Vogelstein. "Neurodata MRI Graphs". *Dockerized python package*. 2022.
3. Jaewon Chung, Benjamin D Pedigo, **Eric W. Bridgeford**, Bijan K Varjavand, Hayden Helm, and Joshua T Vogelstein. "GraSPy: Graph Statistics in Python". *Pypi >1.4M downloads*. May 2018.
4. **Eric W. Bridgeford**, Minh Tang, Jason Yim, and Joshua T Vogelstein. "Linear Optimal Low-Rank Projection". *CRAN > 12,000 downloads*. May 2018.

5. Sambit Panda, Satish Palaniappan, Junhao Xiong, **Eric W. Bridgeford**, Ronak Mehta, Cencheng Shen, and Joshua T. Vogelstein. “Hyppo”. *Pypi ›1.2M downloads*. May 2018.